Tackling Waste Misclassification: Building a Deep Learning System for Smarter Garbage Sorting

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INTRODUCTION

Improper waste disposal remains a major environmental issue. One way to remedy it is by creating a garbage classification system. However, such system is difficult to generalize to the infinite shapes and sizes of objects, and misclassifications can lead to environmental pollution during recycling. Our project leverages deep learning and computer vision to build a robust garbage classification model that can identify waste categories to the highest degree of accuracy.

DATA

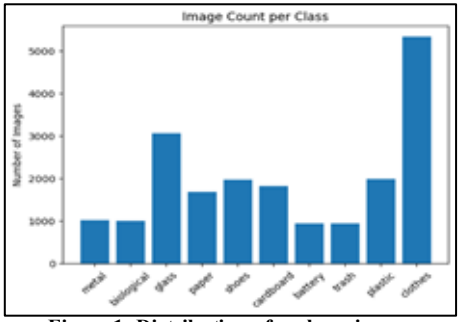


Figure 1: Distribution of garbage images

SYSTEM

The Input: 2D image of an object (jpg)
The Output: 1 of the 10 garbage classes
The Learning Problem:
Supervised learning classification problem

MODEL IMPLEMENTATION

Model Architecture
Our waste classification system is built on a ResNet-50 backbone pre-trained on ImageNet, enhanced with a Convolutional Block Attention Module (CBAM) to improve feature refinement after the backbone output.
CBAM applies channel attention followed by spatial attention, allowing the network to emphasize the most informative feature maps before classification. A custom classifier head replaces the original ResNet FC layer.

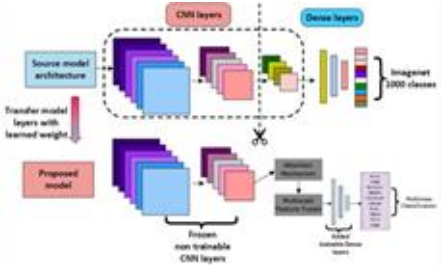


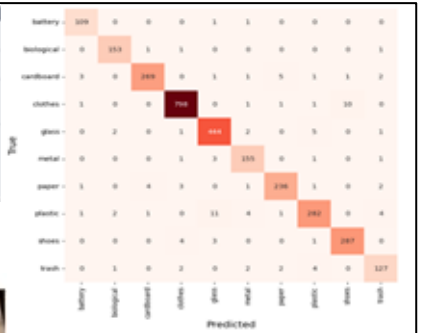
Figure 2: Model Architecture

Two-phase Fine-tuning approach
Phase 1: Backbone + CBAM frozen; only the classifier trained (5 epochs, LR = 0.001) to adapt the new head.
Phase 2: Best Phase 1 model reloaded; deeper backbone layers + CBAM unfrozen and trained with a low LR (0.0001) for up to 15 epochs with early stopping to specialize features.
Optimization: Adam optimizer (lr=1e-3, weight_decay=1e-4) with ReduceLROnPlateau scheduling and early stopping (patience = 5).

RESULTS

Model	Top-1	Top-5	F1 (Macro)
ResNet-50	70.49%	96.49%	61.88%
Vision Transformer	75.30%	97.20%	67.50%
ResNet-50 (Transfer Learning)	96.09%	99.83%	95.15%
ResNet-50 (Transfer Learning, CBAM, Multi Feature Fusion)	**96.56%**	**99.97%**	**95.82%**

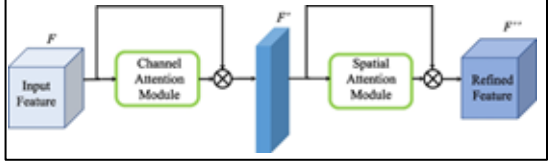
Quantitative comparison across trained models. The ResNet-50 with CBAM and customized classifier achieves the best result in each evaluated metric, as seen in bold.



Qualitative result of best model. Minimal class confusion, making it the most reliable.

NOVELTY

For our novelty, after creating our initial model we added the CBAM after the frozen layers to allow our model to place better precedence for features of importance. This is accomplished through using global averages and maximums. In addition after this we also utilized Multi-Scale Feature Fusion (MSFF). This enables us to utilize all the existing layers in unison, in order to refine the resulting map after CBAM. Finally we are left with a refined map, which as demonstrated in our results, improved our accuracy.



EXCITING ASPECTS OF PROJECT

Technical Aspect: Implementing CBAM attention and multi-scale feature fusion on top ResNet-50 made the model design process engaging

Real-World Impact: Seeing the model achieve such high accuracy in waste classification made the potential impact on reducing contamination and improving recycling feel promising

Performance: Seeing the hybrid architecture outperform baseline models by a large margin made the modeling and training process rewarding